

AutoDirect

Connecting mineral shippers and transporters through one shared job record.

Completed Full-stack project · Freight workflow & coordination

React

Firestore

Transactions

01 / CONTEXT

The problem & my contribution

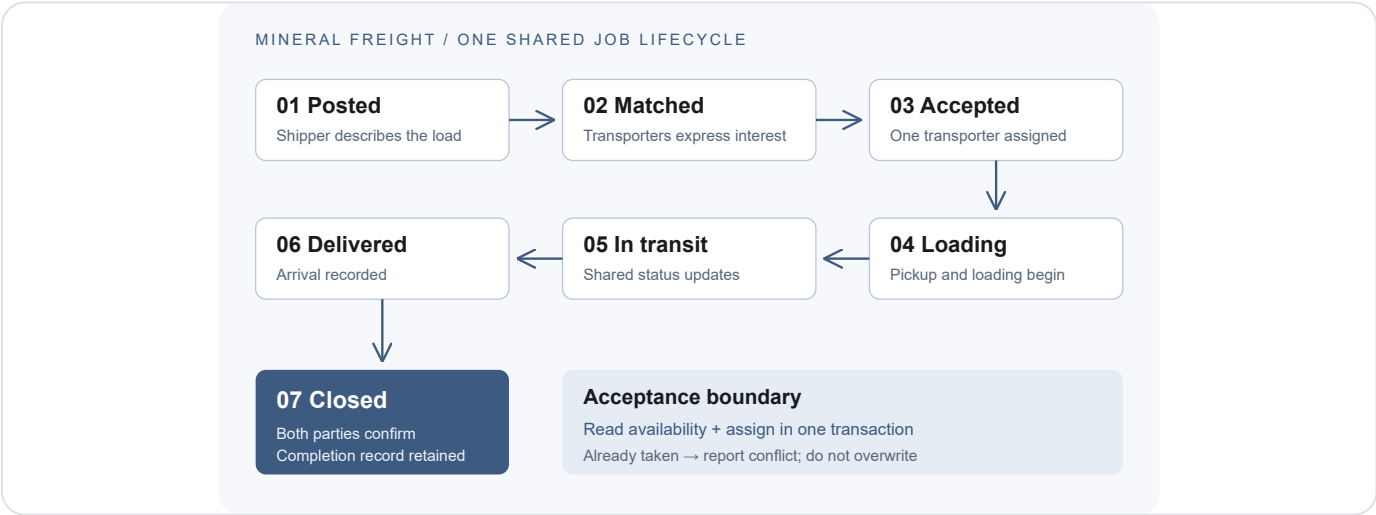
Mineral freight is often arranged through calls, messages, and personal networks. Shippers lack visibility of available capacity, transporters struggle to find return-leg work, and both sides need a shared record of progress. AutoDirect applies a ride-sharing marketplace model to this coordination problem, with shipper and transporter workflows in a React application.

- Keep assignment consistent when multiple transporters accept the same job.
- Represent seven named stages, with clear responsibilities and valid transitions.
- Keep infrastructure lightweight and communicate failures on unreliable connections.

02 / APPROACH

From input to useful output

A shipper posts cargo details, pickup, destination, and timing. Transporters see the request and express interest; one accepts the work. The job then advances through loading, transit, delivery, and closure. Both parties follow the same Firestore record through real-time updates. Acceptance reads availability and writes the assignment in one transaction. If a competing acceptance has already claimed the job, the losing action reports that it is taken rather than overwriting the assignment. Client-side fallbacks provide clear feedback and safe retry paths when the preferred action cannot complete.



Original lifecycle diagram based on the project documentation. Acceptance is the contested transition; final closure follows confirmation by both parties.

How the pieces fit

The documented architecture is a React single-page application communicating directly with Firebase. Firestore stores job records, streams status changes, and provides transactions for assignment. A dedicated custom server is not part of this documented design.

React application	Shipper job posting and tracking; transporter discovery, acceptance, and progress updates.
Firestore	Shared job records, real-time status updates, and transactional assignment.
Client fallbacks	Failure feedback and safe retries when the preferred path cannot finish; not a verified offline-sync system.

Engineering trade-offs

Check and claim atomically Read availability and write assignment in one Firestore transaction. Separate checks and writes leave a race window. Transactional assignment checks shared job state before committing, rather than treating competing requests as independent bookings. <i>Trade-off: Contention and failures need handling, alongside appropriate Firestore rules. The supplied document does not establish those rules.</i>	Make state and failure visible Use named lifecycle stages, a shared record, and explicit failure feedback. Named stages clarify the next action. Shared updates reduce confirmation calls; fallback handling distinguishes saved actions from failed attempts. <i>Trade-off: The managed backend reduces maintenance but depends on Firebase and connectivity. Live GPS tracking remains a proposed extension.</i>
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Results, with boundaries

The supplied documentation reports a complete posting-to-closure workflow, transactional assignment, and graceful failure handling. It provides a concrete design account, but no repository, automated test results, deployment evidence, or measured operating outcomes were supplied for independent verification.

- Seven stages connect marketplace discovery with delivery and a retained completion record.
- The acceptance walkthrough explains the contention case: the first successful claim assigns the transporter; a later claimant receives a taken-job message.

What this does not prove

- Concurrency guarantees, permissions, and retry behaviour still need source-code and test evidence; the document alone does not prove all race conditions are covered.
 - Reduced empty journeys, lower costs, and fewer disputes are intended benefits, not measured pilot results.
- Validate competing acceptance requests, invalid transitions, and network failures. GPS tracking, reputation, pricing suggestions, payments, and shipment documents remain proposed extensions.

Source: AutoDirect portfolio documentation supplied in October 2026. This condensed account distinguishes documented implementation from independent verification and proposed features.